

# Projected Health Impact of Achieving WHO Targets for Cardiovascular Risk-Factor Control

Deterministic sensitivity analysis (one-way scenario analyses)

Work in Progress

WHO · University of Washington

September 2026

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## Study Overview

# Aim 1 Deterministic Sensitivity Analysis

## PILOT RUN (pilot\_Colombia)

1 of 186 countries (Colombia). Software-verification results only: **not global estimates**.

### Question

How robust are the projected deaths averted by the WHO cardiovascular targets to **achievement, timing, statin adherence** and **CVD mortality-trend** assumptions?

### Design (Briggs et al., ISPOR-SMDM 2012)

- One-way deterministic scenarios around the **active base case**
- Every setting re-runs the country-level model (1 country, 6 scenarios, 2017–2050)
- Chosen scenario values, **not** probability distributions: ranges are **not** confidence intervals

### Reference = active base case

BP: 150M additional controlled by 2030 · statins 50% by 2030 · statin adherence IR/CF **0.575 / 0.664** (not 1.0) · CVD CF trend at 80% · background trend on

# DSA Settings

| Setting                        | Family                           | What changes                                    |
|--------------------------------|----------------------------------|-------------------------------------------------|
| Reference                      | Reference                        | —                                               |
| 50% of increment               | Target achievement               | 50% of increment by 2030                        |
| 75% of increment               | Target achievement               | 75% of increment by 2030                        |
| Endpoint 2035                  | Implementation timing            | full endpoint in 2035                           |
| Endpoint 2040                  | Implementation timing            | full endpoint in 2040                           |
| Statin adherence low           | Statin adherence                 | statin adherence 0.50 / 0.65                    |
| Statin adherence moderate      | Statin adherence                 | statin adherence 0.70 / 0.80                    |
| Statin adherence by income     | Statin adherence by income group | statin adherence by income group (illustrative) |
| Perfect adherence (extra ref.) | Extra reference (not base case)  | statin adherence 1.00 / 1.00                    |
| Full CVD CF trend              | CVD mortality trend              | trend_full_cvd                                  |
| No CVD CF trend                | CVD mortality trend              | trend_no_cvd                                    |
| IHME Foresight CVD trend       | CVD mortality trend              | trend_ihme_cvd                                  |

Income-group adherence (L 0.50/0.65, LM 0.60/0.70, UM 0.70/0.80, H 0.80/0.90) is an **illustrative assumption**.  
 Perfect adherence (1.00/1.00) is an extra reference, never the anchor.

## Scenarios

## Four WHO Priority Targets (Reference)

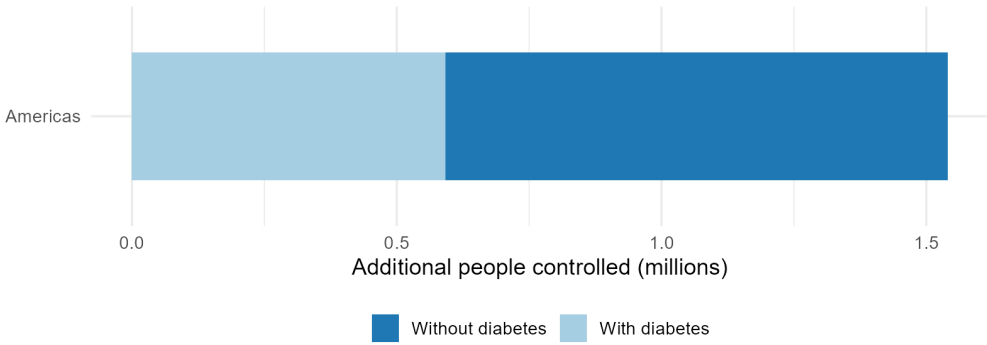
| # | WHO Target                                                    | Modelled Scenario                                                             |
|---|---------------------------------------------------------------|-------------------------------------------------------------------------------|
| 1 | 150M more with HTN controlled by 2030                         | <b>HTN Control (Combined):</b> 1.541M = 0.949M without + 0.592M with diabetes |
| 2 | $\geq 50\%$ eligible adults on lipid-lowering therapy by 2030 | <b>Improved Statin Uptake:</b> 50% coverage by 2030                           |
| 3 | 80% BP control among diagnosed diabetics by 2030              | <b>HTN Control (Diabetes):</b> 0.80 ceiling (full target only)                |
| 4 | 80% availability of validated BP devices                      | Embedded in HTN scenarios                                                     |

The two BP subgroups combine additively; the BP composite combines multiplicatively with statins.



# The 150 Million Split by WHO Region and Diabetes Status

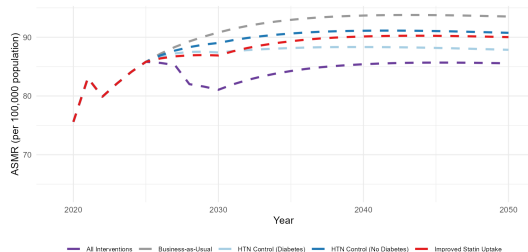
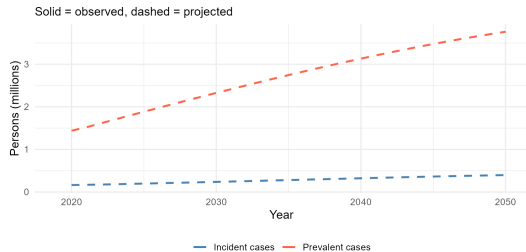
Fixed 2025 population | total 1.5 M



| Region   | HTN pop 2025 (M) | Cov. 2025 | Cov. 2030 | Added (M) |
|----------|------------------|-----------|-----------|-----------|
| Americas | 11.317           | 32.8%     | 46.4%     | 1.541     |
| Global   | 11.317           | 32.8%     | 46.4%     | 1.541     |

## Reference Results

# Baseline Disease Trends and ASMR (Business-as-Usual)



Reference trend regime; solid = calibration, dashed = projection.

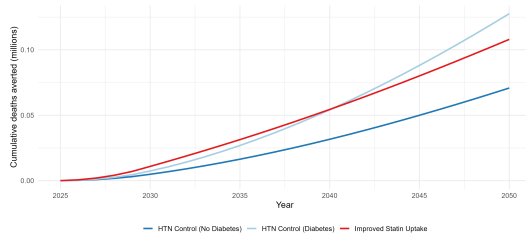
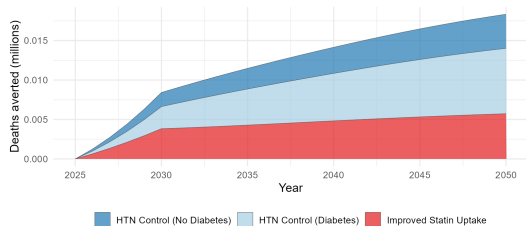
## Summary: Cumulative Deaths Averted, 2025–2050 (Reference)

| Scenario                  | Cumulative (M) | Avg./year (M)  | % of BAU deaths |
|---------------------------|----------------|----------------|-----------------|
| HTN Control (No Diabetes) | 0.071 M        | 0.003 M        | 2.1%            |
| HTN Control (Diabetes)    | 0.128 M        | 0.005 M        | 3.7%            |
| Improved Statin Uptake    | 0.108 M        | 0.004 M        | 3.2%            |
| <b>All Interventions</b>  | <b>0.258 M</b> | <b>0.010 M</b> | <b>7.6%</b>     |

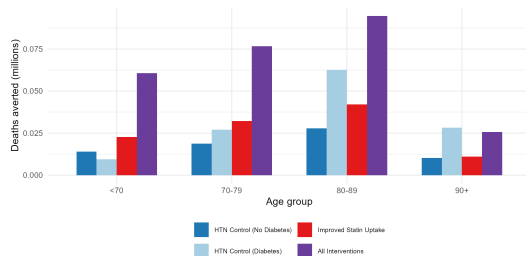
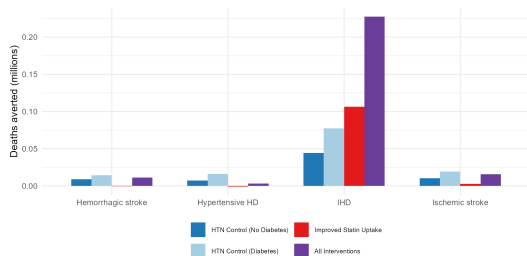
### Headline (reference)

**All Interventions: 0.258 million deaths averted, 2025–2050 ( $\approx$  0.010 M/year).**

# Annual and Cumulative Deaths Averted (Reference)



# Deaths Averted by Cause and Age (Reference)



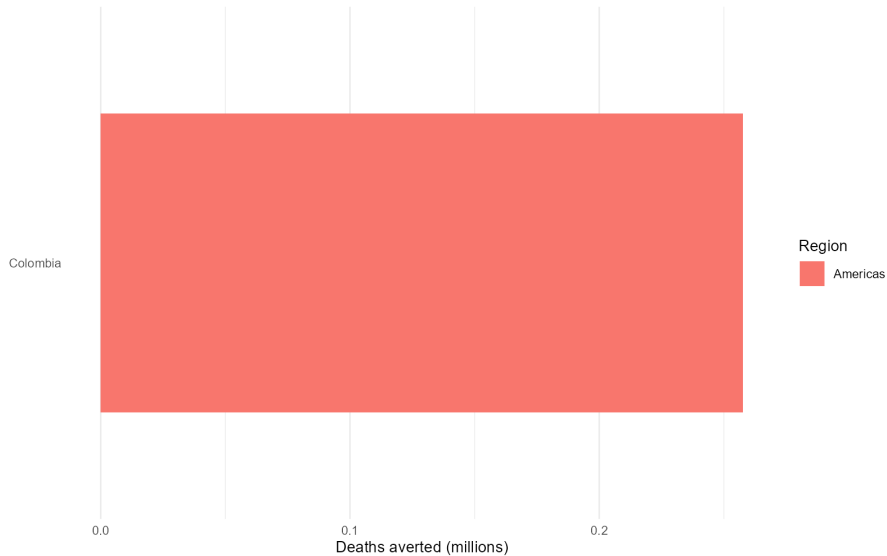
# Burden of Disease Averted and Regional Summary (Reference)

| Intervention              | Deaths (M) | YLLs (M) | YLDs (M) | DALYs (M) |
|---------------------------|------------|----------|----------|-----------|
| HTN Control (No Diabetes) | 0.071      | 0.803    | 0.030    | 0.833     |
| HTN Control (Diabetes)    | 0.128      | 1.046    | 0.058    | 1.103     |
| Improved Statin Uptake    | 0.108      | 1.261    | -0.017   | 1.245     |
| All Interventions         | 0.258      | 3.195    | -0.084   | 3.111     |

| Region   | BAU (M) | Averted (M) | % averted |
|----------|---------|-------------|-----------|
| Americas | 3.404   | 0.258       | 7.6%      |

DALYs 2026–2050 (YLL: WPP 2024 life expectancy; YLD: GBD 2023 disability weights); regions 2025–2050, All Interventions.

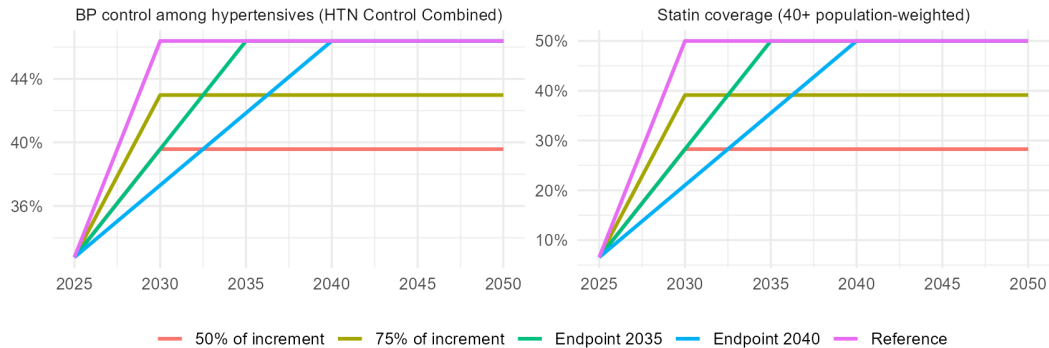
# Top Countries by Projected Impact (Reference)





## Deterministic Sensitivity Analysis

# Coverage Trajectories: Partial Achievement and Delayed Implementation



Adherence and trend settings share the reference trajectories.

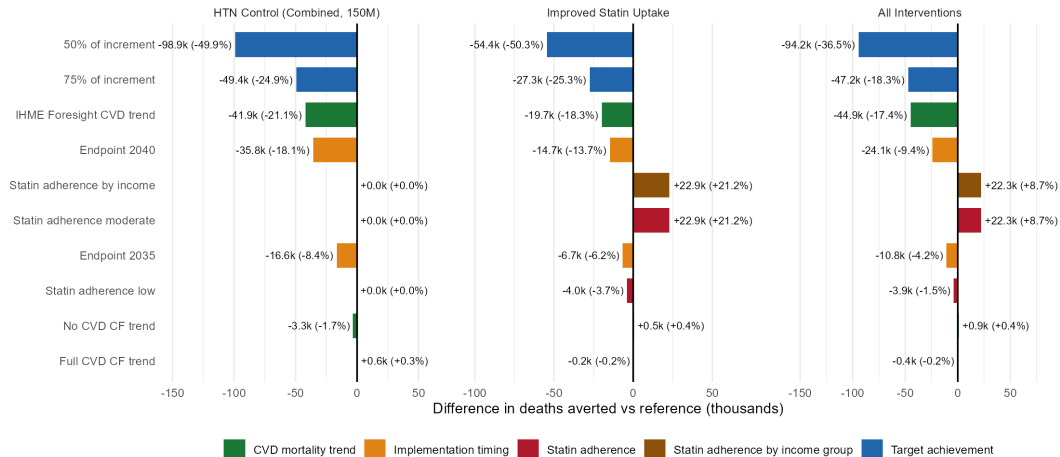
Adherence and trend settings share the reference trajectories. Additional controlled people use the fixed 2025 population of script 04.

# Achieved BP Control and Statin Inputs

| DSA setting      | BP endpoint | Added 2030 (M) | 2035 (M) | 2040 (M) | At endpoint (M) | Share of full target | Diabetes control at endpoint | Statin endpoint | Mean statin baseline / endpoint |
|------------------|-------------|----------------|----------|----------|-----------------|----------------------|------------------------------|-----------------|---------------------------------|
| Reference        | 2030        | 1.54           | 1.54     | 1.54     | 1.54            | 100.0%               | 62.7%                        | 2030            | 6.6% / 50.0%                    |
| 50% of increment | 2030        | 0.77           | 0.77     | 0.77     | 0.77            | 50.0%                | 53.5%                        | 2030            | 6.6% / 28.3%                    |
| 75% of increment | 2030        | 1.16           | 1.16     | 1.16     | 1.16            | 75.0%                | 58.1%                        | 2030            | 6.6% / 39.1%                    |
| Endpoint 2035    | 2035        | 0.77           | 1.54     | 1.54     | 1.54            | 100.0%               | 62.7%                        | 2035            | 6.6% / 50.0%                    |
| Endpoint 2040    | 2040        | 0.51           | 1.03     | 1.54     | 1.54            | 100.0%               | 62.7%                        | 2040            | 6.6% / 50.0%                    |

Partial settings do not reach 150 million: they reach the stated share of the full increment. The diabetic 0.80 figure is the full-target ceiling; the achieved population-weighted diabetes control is shown.

# One-Way Results: Change in Deaths Averted vs Reference



Cumulative 2026–2050; deterministic scenario differences, **not** intervals. Trend settings are compared with their own BAU.

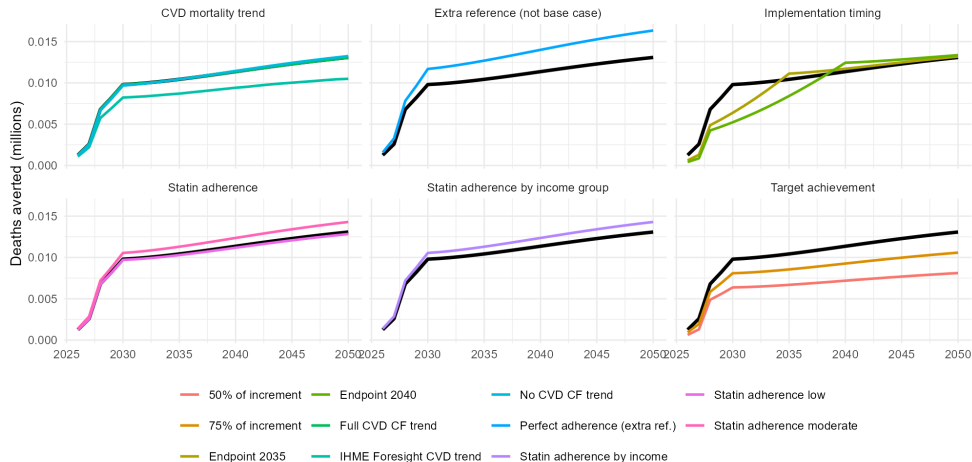
# One-Way Results: All Interventions

| Setting                        | Family                           | Averted (M) | $\Delta$ vs ref. (M) | $\Delta$ (%) | % of BAU |
|--------------------------------|----------------------------------|-------------|----------------------|--------------|----------|
| Reference                      | Reference                        | 0.26        | —                    | —            | 7.7%     |
| 50% of increment               | Target achievement               | 0.16        | -0.09                | -36.5%       | 4.9%     |
| 75% of increment               | Target achievement               | 0.21        | -0.05                | -18.3%       | 6.3%     |
| Endpoint 2035                  | Implementation timing            | 0.25        | -0.01                | -4.2%        | 7.4%     |
| Endpoint 2040                  | Implementation timing            | 0.23        | -0.02                | -9.4%        | 7.0%     |
| Statin adherence low           | Statin adherence                 | 0.25        | -0.00                | -1.5%        | 7.6%     |
| Statin adherence moderate      | Statin adherence                 | 0.28        | +0.02                | +8.7%        | 8.4%     |
| Statin adherence by income     | Statin adherence by income group | 0.28        | +0.02                | +8.7%        | 8.4%     |
| Perfect adherence (extra ref.) | Extra reference (not base case)  | 0.32        | +0.06                | +22.7%       | 9.5%     |
| Full CVD CF trend              | CVD mortality trend              | 0.26        | -0.00                | -0.2%        | 7.7%     |
| No CVD CF trend                | CVD mortality trend              | 0.26        | +0.00                | +0.4%        | 7.8%     |
| IHME Foresight CVD trend       | CVD mortality trend              | 0.21        | -0.04                | -17.4%       | 8.5%     |

## Range across one-way settings (All Interventions, 2026–2050)

0.164 M (50% of increment) to 0.280 M (Statin adherence moderate) vs 0.258 M in the reference — a deterministic spread, not a confidence interval.

# Annual Deaths Averted by DSA Setting



# Endpoints: 2030, 2035, 2040, 2050 (All Interventions)

| DSA setting                    | Annual 2030 | Annual 2035 | Annual 2040 | Annual 2050 | Cumulative 2026–2030 | Cumulative 2026–2035 | Cumulative 2026–2040 | Cumulative 2026–2050 |
|--------------------------------|-------------|-------------|-------------|-------------|----------------------|----------------------|----------------------|----------------------|
| Reference                      | 0.010       | 0.010       | 0.011       | 0.013       | 0.029                | 0.079                | 0.134                | 0.258                |
| 50% of increment               | 0.006       | 0.007       | 0.007       | 0.008       | 0.019                | 0.051                | 0.086                | 0.164                |
| 75% of increment               | 0.008       | 0.009       | 0.009       | 0.011       | 0.024                | 0.065                | 0.110                | 0.210                |
| Endpoint 2035                  | 0.006       | 0.011       | 0.012       | 0.013       | 0.019                | 0.064                | 0.122                | 0.247                |
| Endpoint 2040                  | 0.005       | 0.008       | 0.012       | 0.013       | 0.015                | 0.051                | 0.105                | 0.234                |
| Statin adherence low           | 0.010       | 0.010       | 0.011       | 0.013       | 0.028                | 0.078                | 0.133                | 0.254                |
| Statin adherence moderate      | 0.011       | 0.011       | 0.012       | 0.014       | 0.031                | 0.086                | 0.145                | 0.280                |
| Statin adherence by income     | 0.011       | 0.011       | 0.012       | 0.014       | 0.031                | 0.086                | 0.145                | 0.280                |
| Perfect adherence (extra ref.) | 0.012       | 0.013       | 0.014       | 0.016       | 0.034                | 0.095                | 0.163                | 0.316                |
| Full CVD CF trend              | 0.010       | 0.010       | 0.011       | 0.013       | 0.029                | 0.079                | 0.134                | 0.257                |
| No CVD CF trend                | 0.010       | 0.010       | 0.011       | 0.013       | 0.028                | 0.079                | 0.134                | 0.259                |
| IHME Foresight CVD trend       | 0.008       | 0.009       | 0.009       | 0.010       | 0.024                | 0.067                | 0.112                | 0.213                |

Deaths averted (millions): annual in the endpoint year and cumulative from 2026.

# Statin Adherence by World Bank Income Group (Illustrative)

| Intervention           | Income group        | Reference | Income-group adherence | $\Delta$ (M) | $\Delta$ (%) |
|------------------------|---------------------|-----------|------------------------|--------------|--------------|
| All Interventions      | Upper-middle income | 0.258     | 0.280                  | +0.022       | +8.7%        |
| All Interventions      | Global              | 0.258     | 0.280                  | +0.022       | +8.7%        |
| Improved Statin Uptake | Upper-middle income | 0.108     | 0.131                  | +0.023       | +21.2%       |
| Improved Statin Uptake | Global              | 0.108     | 0.131                  | +0.023       | +21.2%       |

Adherence changes statin incidence (IR) and case-fatality (CF) effects only; BP-only scenarios are unchanged (verified for every country).

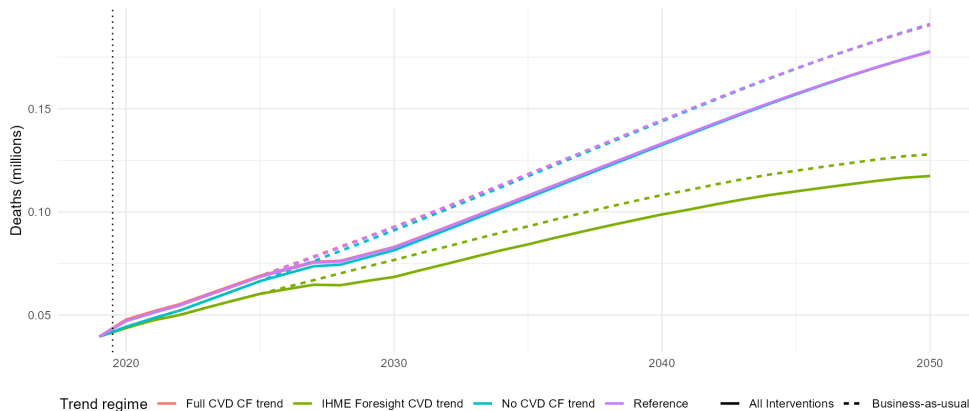


# Alternative CVD Mortality Trends (Structural)

| Mortality trend          | Intervention                 | Projected deaths (M) | BAU deaths (M) | Deaths averted (M) | Averted per year (M) | Averted (% of BAU) | Δ averted vs reference (M) | Δ (%)  |
|--------------------------|------------------------------|----------------------|----------------|--------------------|----------------------|--------------------|----------------------------|--------|
| Reference                | HTN Control (Combined, 150M) | 3.14                 | 3.34           | 0.20               | 0.008                | 5.9%               | —                          | —      |
| Reference                | Improved Statin Uptake       | 3.23                 | 3.34           | 0.11               | 0.004                | 3.2%               | —                          | —      |
| Reference                | All Interventions            | 3.08                 | 3.34           | 0.26               | 0.010                | 7.7%               | —                          | —      |
| Full CVD CF trend        | HTN Control (Combined, 150M) | 3.14                 | 3.34           | 0.20               | 0.008                | 6.0%               | +0.00                      | +0.3%  |
| Full CVD CF trend        | Improved Statin Uptake       | 3.23                 | 3.34           | 0.11               | 0.004                | 3.2%               | -0.00                      | -0.2%  |
| Full CVD CF trend        | All Interventions            | 3.08                 | 3.34           | 0.26               | 0.010                | 7.7%               | -0.00                      | -0.2%  |
| No CVD CF trend          | HTN Control (Combined, 150M) | 3.12                 | 3.32           | 0.19               | 0.008                | 5.9%               | -0.00                      | -1.7%  |
| No CVD CF trend          | Improved Statin Uptake       | 3.21                 | 3.32           | 0.11               | 0.004                | 3.3%               | +0.00                      | +0.4%  |
| No CVD CF trend          | All Interventions            | 3.06                 | 3.32           | 0.26               | 0.010                | 7.8%               | +0.00                      | +0.4%  |
| IHME Foresight CVD trend | HTN Control (Combined, 150M) | 2.34                 | 2.50           | 0.16               | 0.006                | 6.3%               | -0.04                      | -21.1% |
| IHME Foresight CVD trend | Improved Statin Uptake       | 2.41                 | 2.50           | 0.09               | 0.004                | 3.5%               | -0.02                      | -18.3% |
| IHME Foresight CVD trend | All Interventions            | 2.29                 | 2.50           | 0.21               | 0.009                | 8.5%               | -0.04                      | -17.4% |

Background trend on in every regime; each regime compared with its own BAU. Layout after Table S8 of the related RTSL draft.

# CVD Deaths by Trend Regime: BAU and All Interventions



# Burden of Disease Averted by DSA Setting

| DSA setting                    | Deaths (M) | YLLs (M) | YLDs (M) | DALYs (M) | Δ DALYs vs ref. (M) | Δ (%)  |
|--------------------------------|------------|----------|----------|-----------|---------------------|--------|
| Reference                      | 0.26       | 3.19     | -0.08    | 3.11      | —                   | —      |
| 50% of increment               | 0.16       | 2.13     | -0.06    | 2.07      | -1.04               | -33.5% |
| 75% of increment               | 0.21       | 2.66     | -0.07    | 2.59      | -0.52               | -16.7% |
| Endpoint 2035                  | 0.25       | 3.04     | -0.08    | 2.97      | -0.14               | -4.6%  |
| Endpoint 2040                  | 0.23       | 2.87     | -0.07    | 2.80      | -0.31               | -10.1% |
| Statin adherence low           | 0.25       | 3.15     | -0.09    | 3.07      | -0.04               | -1.4%  |
| Statin adherence moderate      | 0.28       | 3.45     | -0.09    | 3.36      | +0.25               | +8.0%  |
| Statin adherence by income     | 0.28       | 3.45     | -0.09    | 3.36      | +0.25               | +8.0%  |
| Perfect adherence (extra ref.) | 0.32       | 3.85     | -0.09    | 3.76      | +0.65               | +21.0% |
| Full CVD CF trend              | 0.26       | 3.19     | -0.08    | 3.11      | -0.00               | -0.1%  |
| No CVD CF trend                | 0.26       | 3.20     | -0.08    | 3.12      | +0.01               | +0.2%  |
| IHME Foresight CVD trend       | 0.21       | 2.44     | -0.06    | 2.38      | -0.73               | -23.6% |

All Interventions, 2026–2050, each setting against its own BAU. Negative YLDs averted: people kept alive live longer with CVD (as in the ordinary report).

# Economic Value by DSA Setting

| DSA setting                    | VSL (USD T) | VSL % GNI | VSLY (USD T) | VSLY % GNI | Δ VSLY vs ref. |
|--------------------------------|-------------|-----------|--------------|------------|----------------|
| Reference                      | 0.61        | 2.3%      | 0.24         | 0.9%       | —              |
| 50% of increment               | 0.39        | 1.4%      | 0.16         | 0.6%       | -33.7%         |
| 75% of increment               | 0.50        | 1.9%      | 0.20         | 0.7%       | -16.9%         |
| Endpoint 2035                  | 0.59        | 2.2%      | 0.23         | 0.8%       | -4.3%          |
| Endpoint 2040                  | 0.56        | 2.1%      | 0.21         | 0.8%       | -9.2%          |
| Statin adherence low           | 0.60        | 2.2%      | 0.23         | 0.9%       | -1.3%          |
| Statin adherence moderate      | 0.66        | 2.5%      | 0.25         | 0.9%       | +8.0%          |
| Statin adherence by income     | 0.66        | 2.5%      | 0.25         | 0.9%       | +8.0%          |
| Perfect adherence (extra ref.) | 0.75        | 2.8%      | 0.28         | 1.1%       | +20.8%         |
| Full CVD CF trend              | 0.61        | 2.3%      | 0.24         | 0.9%       | -0.1%          |
| No CVD CF trend                | 0.61        | 2.3%      | 0.24         | 0.9%       | +0.2%          |
| IHME Foresight CVD trend       | 0.51        | 1.9%      | 0.18         | 0.7%       | -23.7%         |

All Interventions, World, cumulative 2026–2050; VSL/VSLY primary elasticity (0.8/1.2), discounted at 3% to 2026; recomputed from each setting's own deaths averted.

## Economic Value of Avoidable Mortality (Reference)

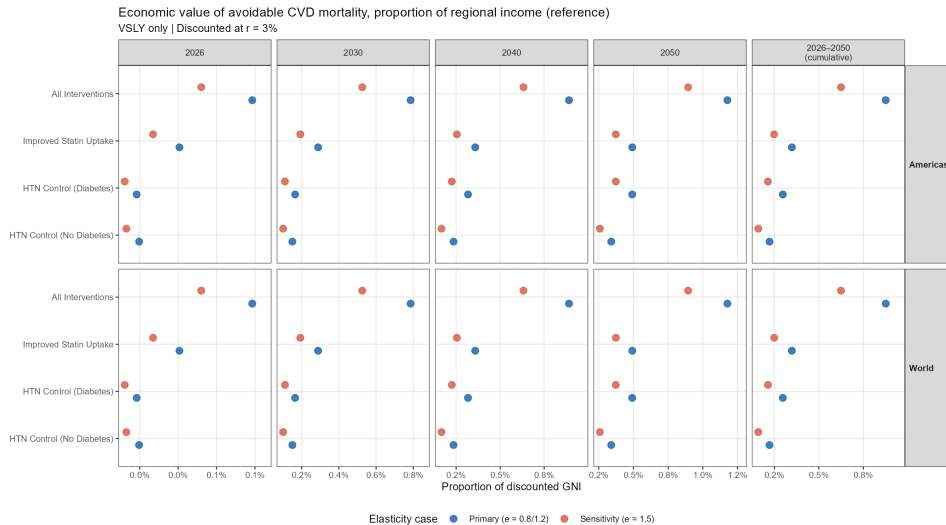
# Economic Value by WHO Region (Primary, Cumulative 2026–2050)

World: All Interventions (reference, primary, 3%)

**VSLY = 0.9%** of annual income PPP (cumulative **\$0.2 T**).

| Region   | VSL (USD T) | VSL % GNI | VSLY (USD T) | VSLY % GNI |
|----------|-------------|-----------|--------------|------------|
| Americas | 0.61        | 2.3%      | 0.24         | 0.9%       |
| World    | 0.61        | 2.3%      | 0.24         | 0.9%       |

# Value of Avoidable Mortality as a Proportion of Income (Reference)



## Discussion



# Interpretation

## What the one-way settings show

- Target achievement (50%/75% of the increment) scales benefits roughly in proportion to the achieved increment
- Delaying the full endpoint to 2035/2040 loses early benefits; effects converge once the endpoint is reached
- Statin adherence moves statin-driven benefits only; BP benefits are unchanged
- The CVD case-fatality trend regime changes both BAU and intervention deaths; compare within regimes

## Interpretation rule

Deterministic scenario spreads describe the consequences of named assumptions; they are not probability statements or confidence intervals.

# Limitations of the DSA

- One-way settings only; no multi-way stress tests or probability weights
- Income-group statin adherence values are illustrative (no sourced income-specific estimate in the repository)
- No separate BP-medication adherence: BP targets are achieved control, which already embodies adherence
- The cause-specific background-mortality trend input has no post-2019 values, so only the all-cause envelope (BG.mx.all) trends, in every regime alike
- The reference CVD trend series starts above the 2019 level for IHD and ischemic stroke; the IHME Foresight trend is global, cause-level and age-standardised
- Active statin CF adherence 0.664 vs 64.4% reported for secondary prevention by Basios et al. 2025 (kept to reproduce the pipeline)

# Validation

| Result        | n  |
|---------------|----|
| pass          | 99 |
| informational | 14 |

Checks by code/10\_deterministic\_dsa.R (run pilot\_Colombia): DSA reference bit-identical to the live 06 functions and to the ordinary 06 output; 05 reproduced for every trend regime; trajectories, targets, adherence and trend formulas recomputed independently.

## Appendix

# Notation: Core Model and HTN Control

| Symbol                                   | Description                                                                                   |
|------------------------------------------|-----------------------------------------------------------------------------------------------|
| $W, K, D$                                | Well, Sick, Dead populations                                                                  |
| $IR, CF$                                 | Baseline incidence & case-fatality rates                                                      |
| $\varepsilon_{IR,j}, \varepsilon_{CF,j}$ | Multiplicative effect ratios for intervention $j$                                             |
| $BG.mx, BG.mx.all$                       | Cause-specific & all-cause background mortality                                               |
| $c_{g,0}, c_{g,t}, c_{g,target}$         | Baseline, time- $t$ , target control, subgroup $g \in \{N, D\}$                               |
| $f$                                      | DSA fraction of the increment: $c_{target}^{DSA} = c_0 + f(c_{target} - c_0)$                 |
| $T$                                      | Endpoint year: $c_t = c_0 + (c_{target} - c_0) \min[\max(0, (t - 2025)/(T - 2025)), 1]$       |
| $e_{g,t}$                                | $E_g(c_{g,t} - c_{g,0}) / (1 - E_g c_{g,0})$                                                  |
| $w_D$                                    | Diabetes share among hypertensives; $\varepsilon_{IR,BP} = 1 - (1 - w_D)e_{N,t} - w_De_{D,t}$ |

# Statin Therapy and Adherence

*Incremental coverage above baseline  $U_{0,\ell}$  (DSA endpoint  $T$ ):*

$$\Delta U(t, \ell) = \Delta U_{\text{target}, \ell} \min \left[ \frac{t - 2026 + 1}{T - 2026 + 1}, 1 \right]$$

$$\Delta U_{\text{target}, \ell} = \max(U_{\text{target}}^{DSA} - U_{0, \ell}, 0),$$

$$U_{\text{target}}^{DSA} = U_0 + f(0.50 - U_0)$$

*Adherence  $a_{IR}$ ,  $a_{CF}$  enter both numerator and baseline denominator:*

$$E^{IR} = \frac{AF \delta^{IR} \Delta U a_{IR}}{1 - \delta^{IR} U_0 a_{IR}}, \quad E^{CF} = \frac{\delta^{CF} \Delta U a_{CF}}{1 - \delta^{CF} U_0 a_{CF}}$$

Reference  $a_{IR} = 0.575$ ,  $a_{CF} = 0.664$ .

# References

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